

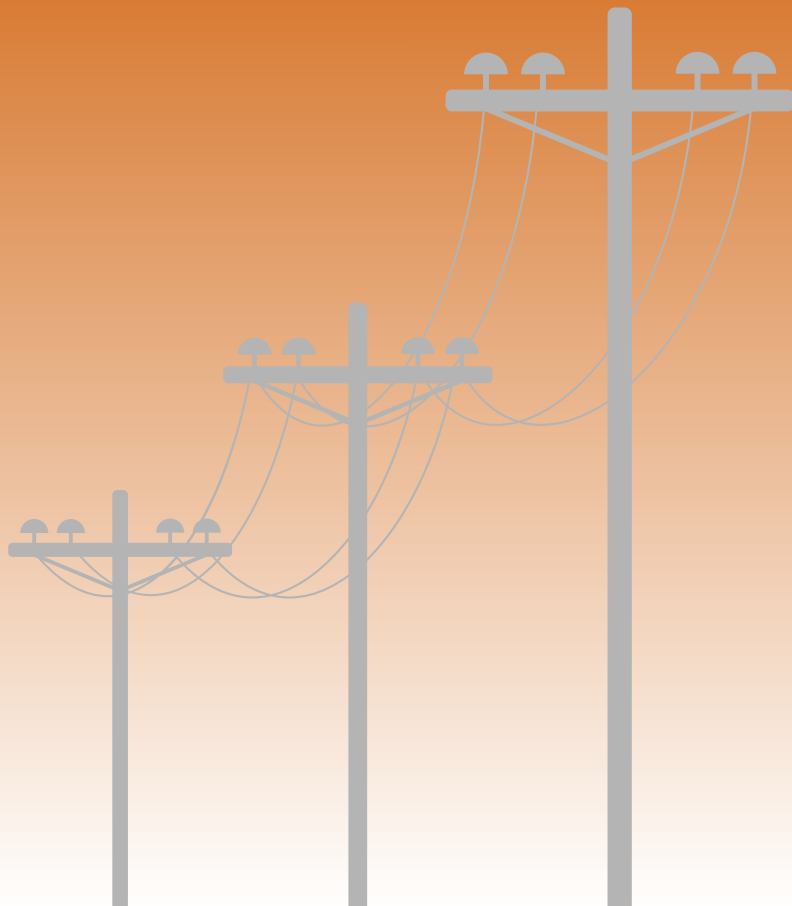
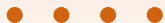


Power Outages



EDISON
INTERNATIONAL®

x



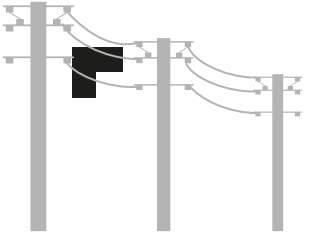


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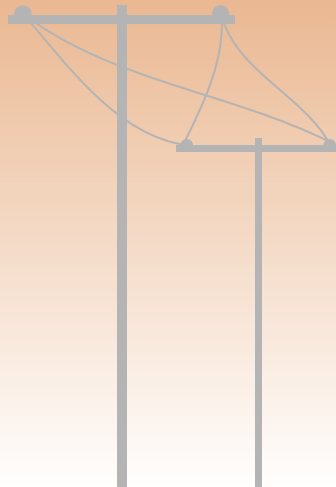
06

Recommendations



01

Background





Negative Impact of Power Outages

Black-outs cause significant

Health Risks

(Lack of heating & cooling, interruption to healthcare facilities, patients in critical care, food and water safety)

One black out can cost the U.S Economy

\$150 billion

annually

("Bloom")

Black-outs risk

Safety and Security

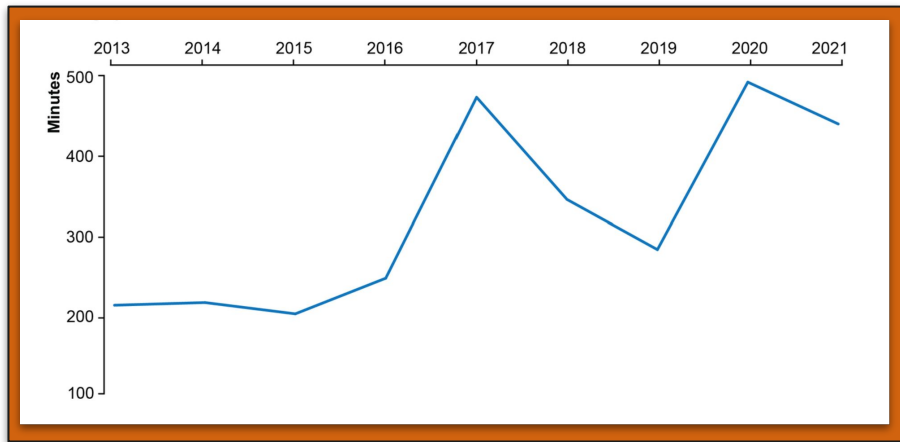
(Impair lighting and security systems, increasing risks of accidents and crime.)



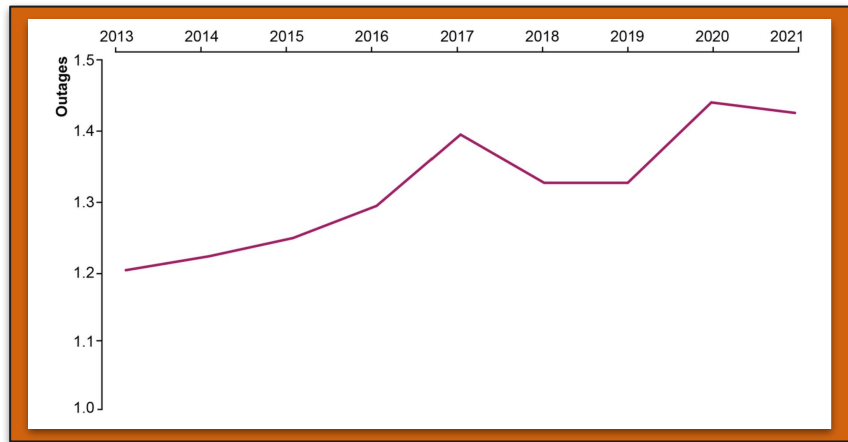


Power Outage Trends **Over the Years**

Average **Duration** of a Power Outage

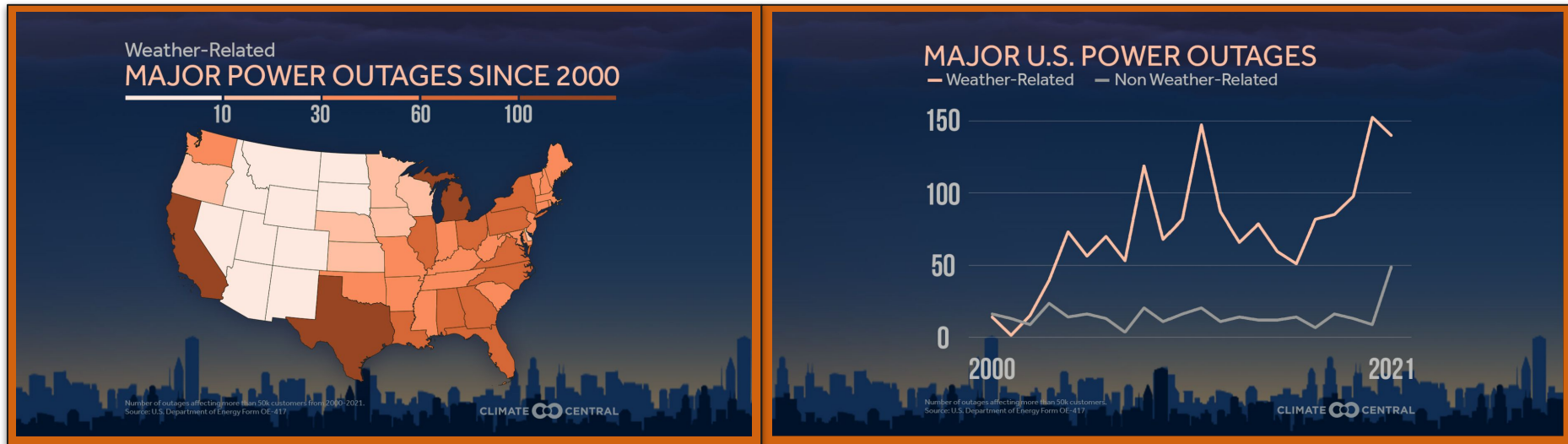


Average **Number** of Outages for a Customer





Weather-Related Power Outages



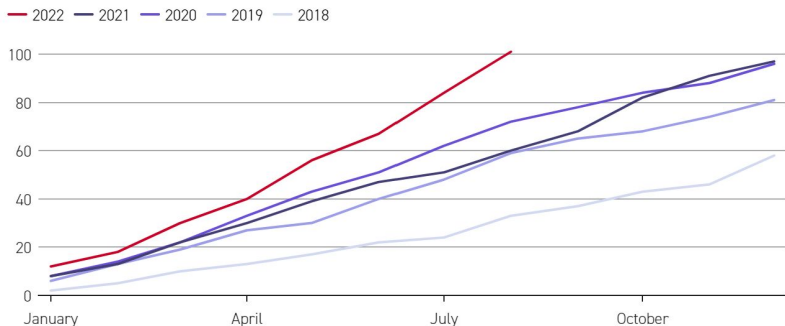
“Between 2011 and 2021, there was a **significant increase** in the average annual number of weather-related power outages across the U.S., and similar trends have been observed in California (“Surging”, 2022)”.



Vandalism-Related Power Outages

Power grid attacks are on the rise this year

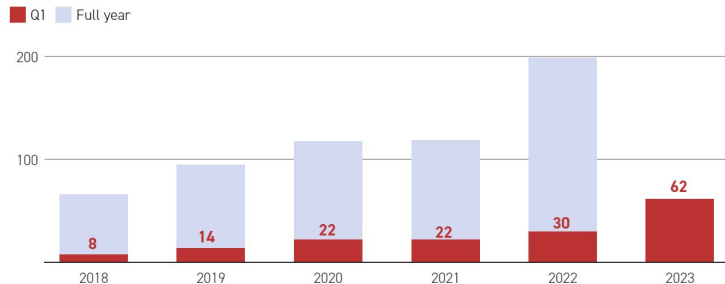
Cumulative number of reported human-caused attacks on power grid infrastructure in the past five years



Source: DOE
Catherine Morehouse / POLITICO

Attacks and threats against the grid doubled in the first quarter of 2023

Incidents that utilities reported to the Energy Department. Most recent data is from January-March of this year.



Source: DOE
Catherine Morehouse/POLITICO

“Grid attacks that led to power outages increased **71 percent** from 2021 to 2022. That increase was primarily due to a rise in gunfire assaults against critical infrastructure (Morehouse, 2023).”



SWOT ANALYSIS



TRENGTHS

- High investment in power outage prevention
- Planned maintenances to spot potential vulnerabilities
- Advanced data collection methods



WEAKNESSES

- Aging infrastructure
- Customer satisfaction issues
 - *Net Promoter Score* is -28 (on a scale from -100 to 100)



OPPORTUNITIES

- Technological Advancements
 - Predictive analytics
 - Advanced weather forecasting
- Grid Hardening and Resilience
- Enhanced Security and Surveillance



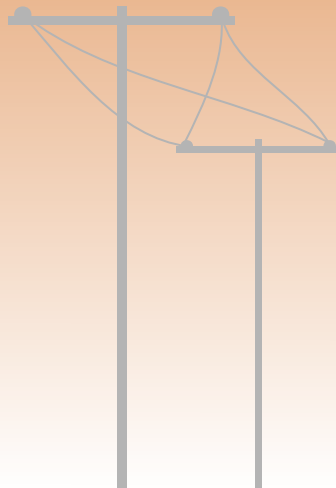
THREATS

- Climate Change
 - Increase in severe weather
 - Wildfires, rainstorms, floods
- Increase in crime
 - Vandalism
 - Damage to power lines and transformers



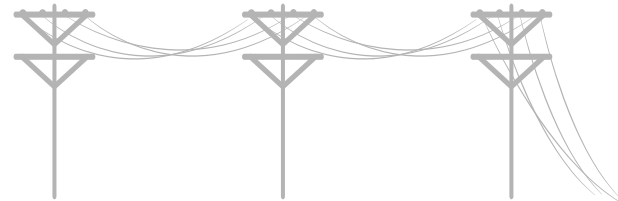
02

Problem Statement





Problem Statement



Southern California Edison must address the escalating frequency of power outages, intensified by severe heat waves, rainstorms, and rising instances of vandalism.

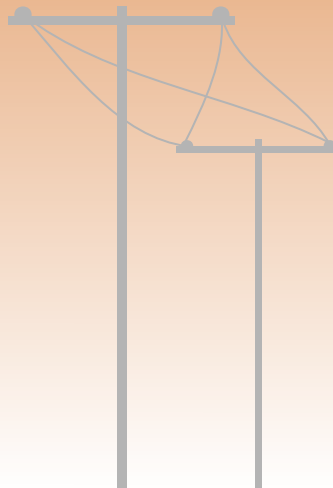
Our goal is to leverage machine learning and data analysis to enable SoCal Edison to rapidly identify the outage type and expected duration of a power outage.





03

Severe Weather



Weather-Related Power Outages



UTUMN

- Strong Winds (Santa Ana Winds)
- Wildfire Risk



INTER

- Severe Storms
- Landslides
- Increase of Natural Gas Prices / Shortage of Natural Gas



UMMER

- Extreme Heat
- Wildfires
- Dry Air



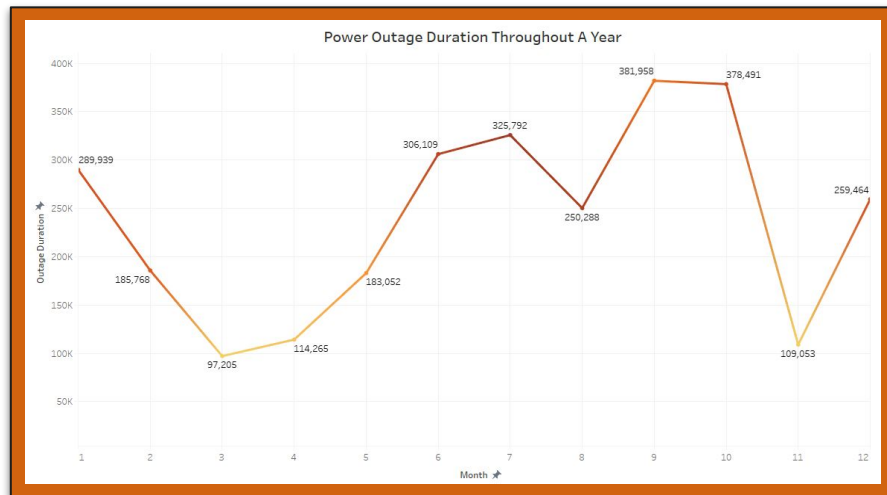
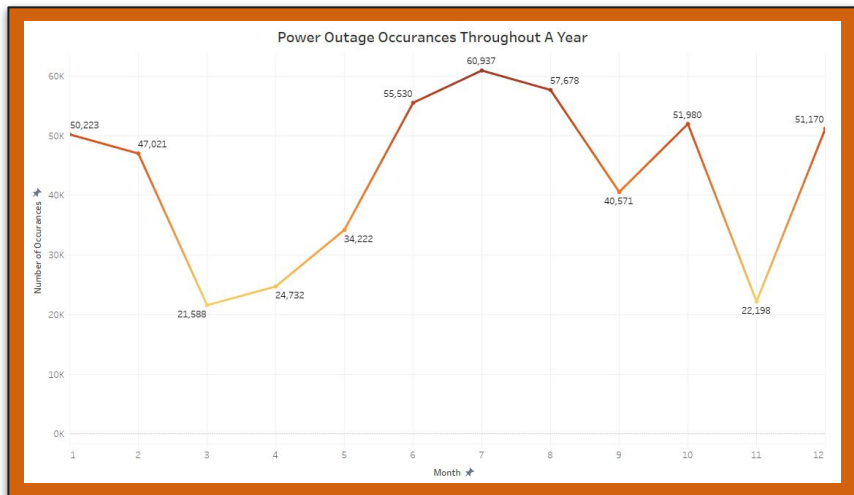
PRING

- Moderate Rains
- Brush Fire Risk
- Temperature Fluctuations



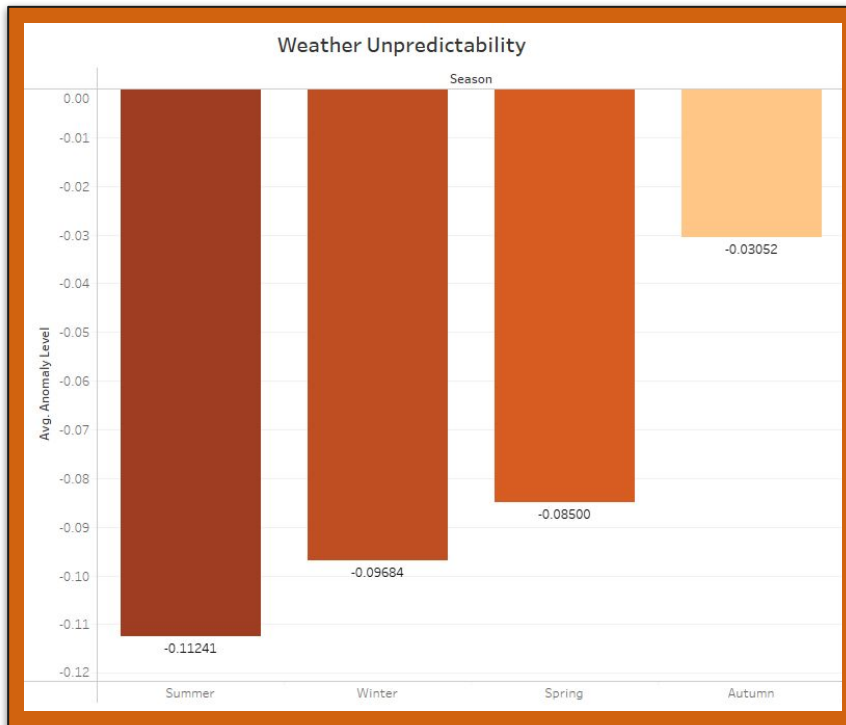
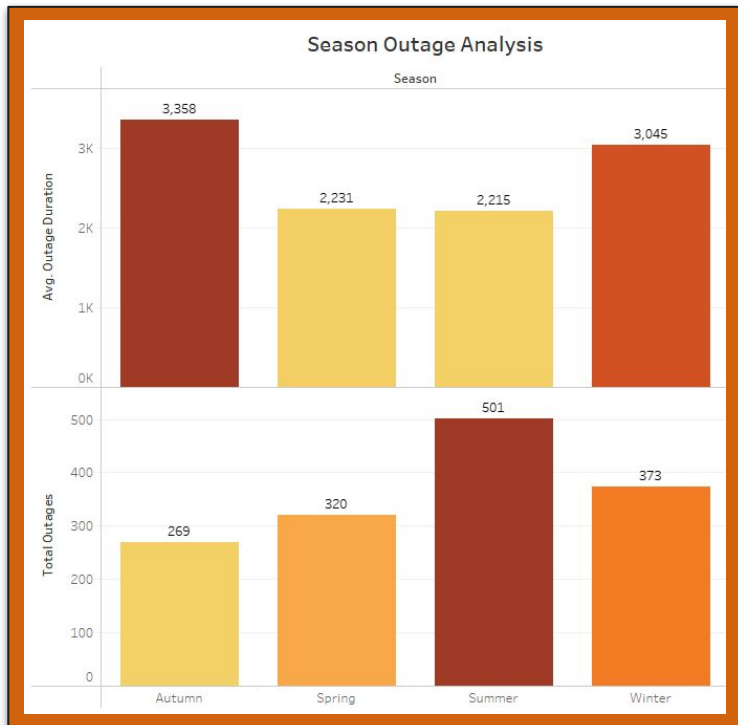


Outages Throughout a Year



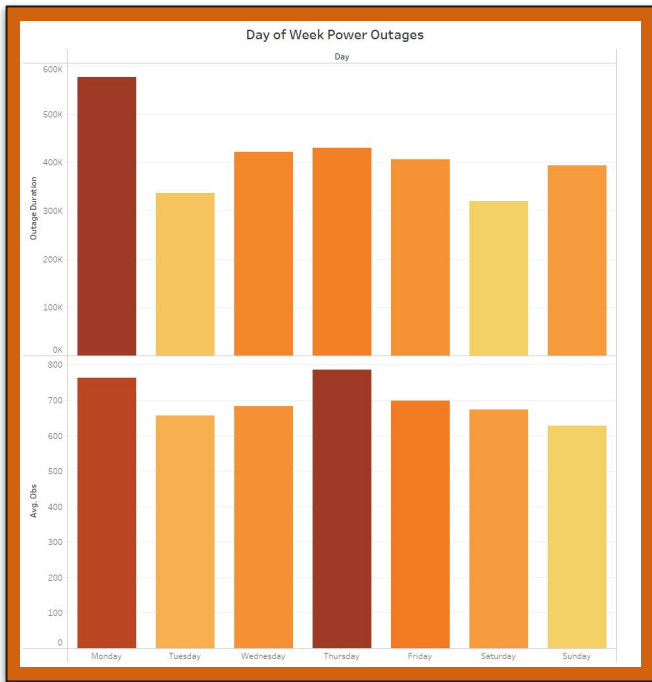


Seasonal Analysis





Outages Throughout a Week



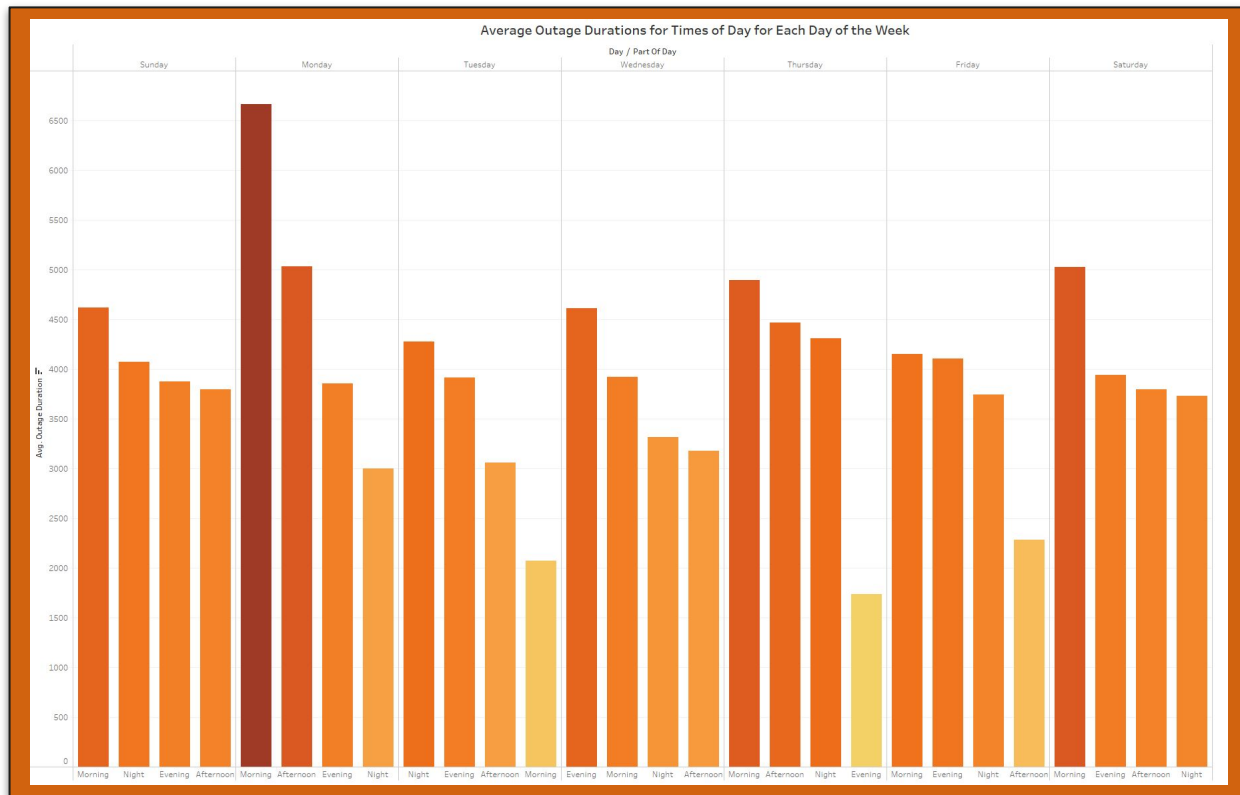
Outage Duration Heatmap

Season	Day						
	Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
Autumn	5,680	5,268	5,717	5,618	6,120	4,567	7,944
Spring	2,555	4,264	3,496	3,653	2,909	2,233	3,794
Summer	2,961	5,299	2,968	2,570	2,526	3,051	2,713
Winter	4,473	3,287	2,665	2,939	4,682	4,583	4,246





Outages Throughout a Day/Week





Yearly/Seasonal Key Insights



Most Severe



Fall
(Sep-Oct)

Higher Number of Outages
Longest Outage Durations
Most Predictable

Most Frequent and
Unpredictable



**Summer
& Winter**



Highest Number of Outages
Shorter Outage Durations
Least Predictable



Least Affected

Spring



Low of Outages
Shortest Outage Durations
Average Predictability



Severity = Outage Duration
Risk = Number of Outages

Key Insights by Day/Week

Highest Risk and
Severity

Monday

Highest Number of Outages
Highest Severity



Low Risk

**Tuesday
&
Saturday**

Low Number of Outages
Lowest Severity



Most Severe

Thursday

Longest Total Outage Duration
High Number of Outages

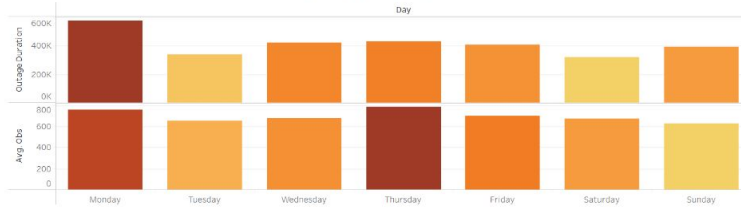


Severe Weather Dashboard

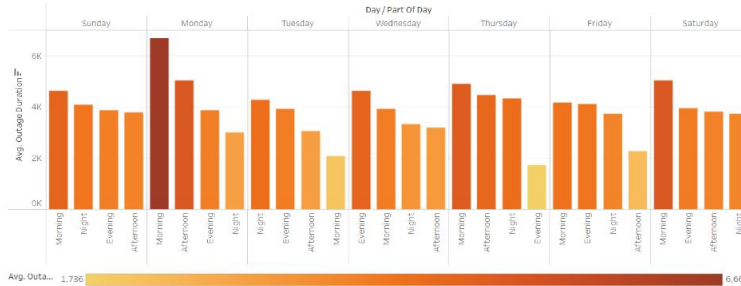
Outage Duration Heatmap



Day of Week Power Outages



Average Outage Durations for Times of Day for Each Day of the Week



Power Outages Throughout A Year



Power Outage Duration Through A Year



Weather Unpredictability



Click the image to view the full dashboard

Our Predictive Model

Logistic Regression Model

What does it do?

Predicts whether an outage was caused by **severe weather** or **not**.

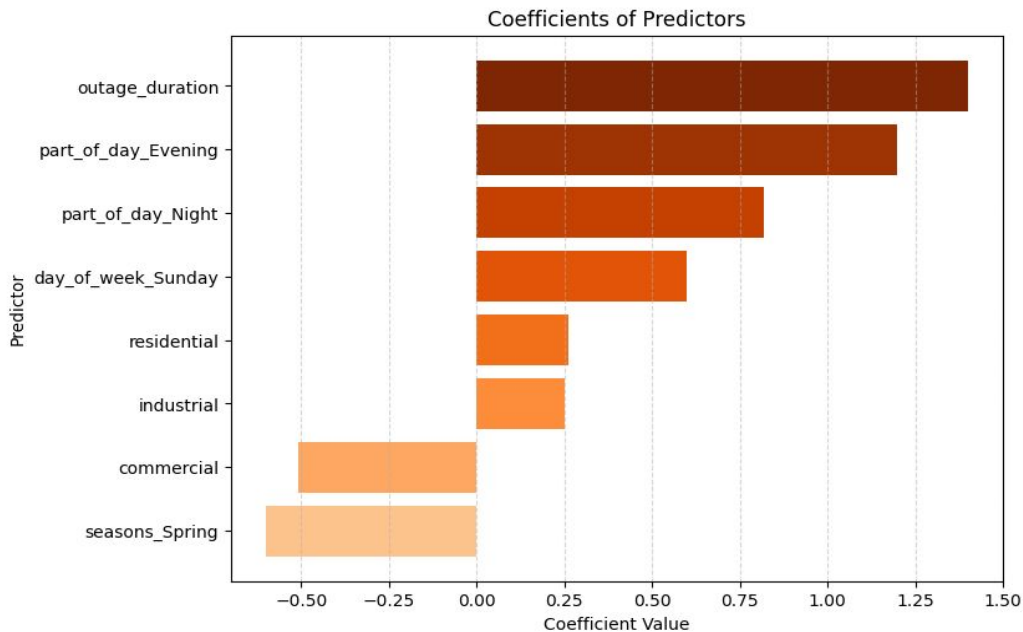
- Probability ≥ 0.5
 - Classification: Severe Weather
- Probability < 0.5
 - Classification: Not Severe Weather

How does it work?

Calculates probability based on feature coefficients for any given outage.

$$h\theta(X) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X)}}$$

Logistic Regression Model - Coefficients



76% accuracy

Demo

A power outage has occurred and you want to know if it was caused by severe weather

Features

- INTERCEPT:
 - 0.00584248
- Outage Duration
 - 3
- Night
 - YES (1)
- Commercial
 - -2
- Residential
 - 2.4
- Industrial
 - 0.3
- Sunday
 - YES (1)
- Spring
 - YES (1)

Calculation

$$p = \frac{1}{1 + e^{-7.12584248}}$$

=

0.999221562009

=

There is a **99.92% chance**
this power outage was due
to **severe weather**

Action Plans

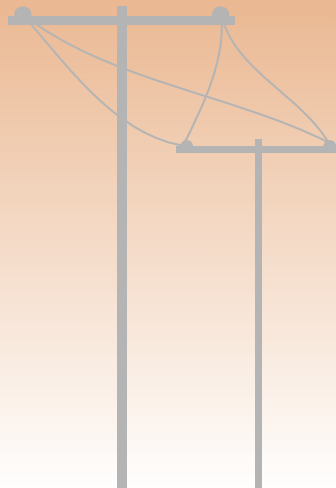
- **Strengthen** grid infrastructure like transformers, power lines, and other components.
- **Improve** weather forecasting and response time to existing outages.
- **Educate** the general public on the potential outages posed by severe weather.





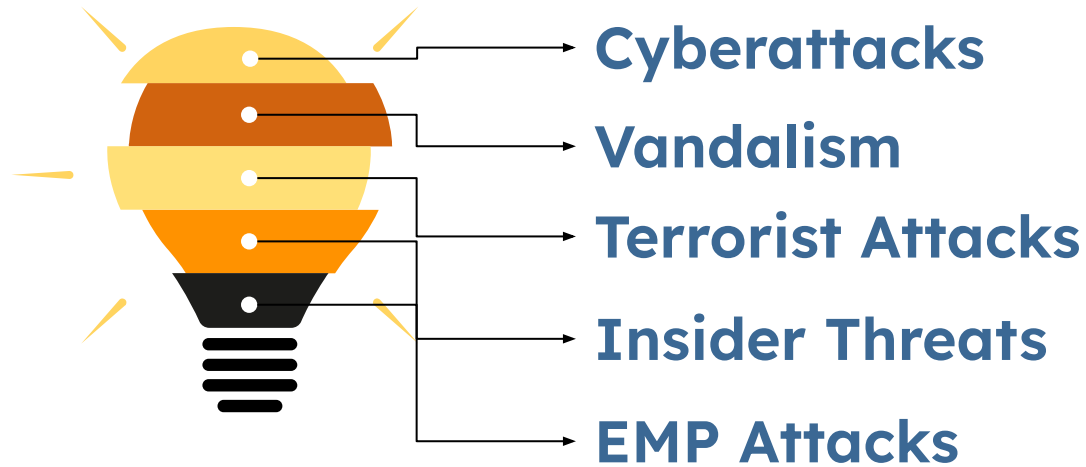
04

Intentional Attacks



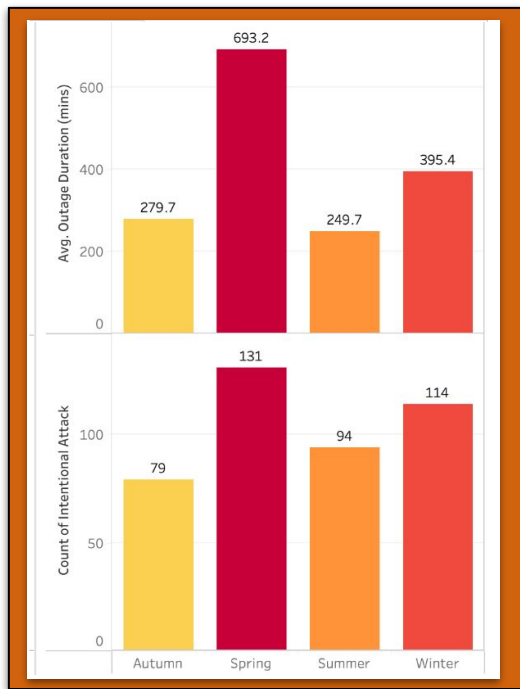


Types of Intentional Attacks

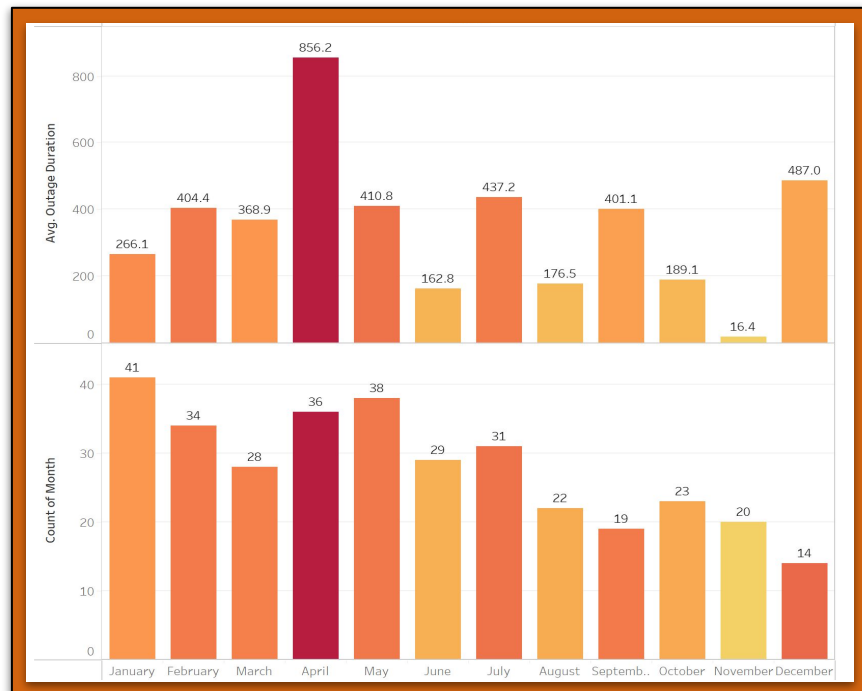


Number of Outages and Duration Time Throughout a Year

Season

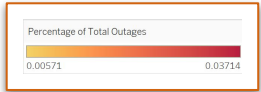
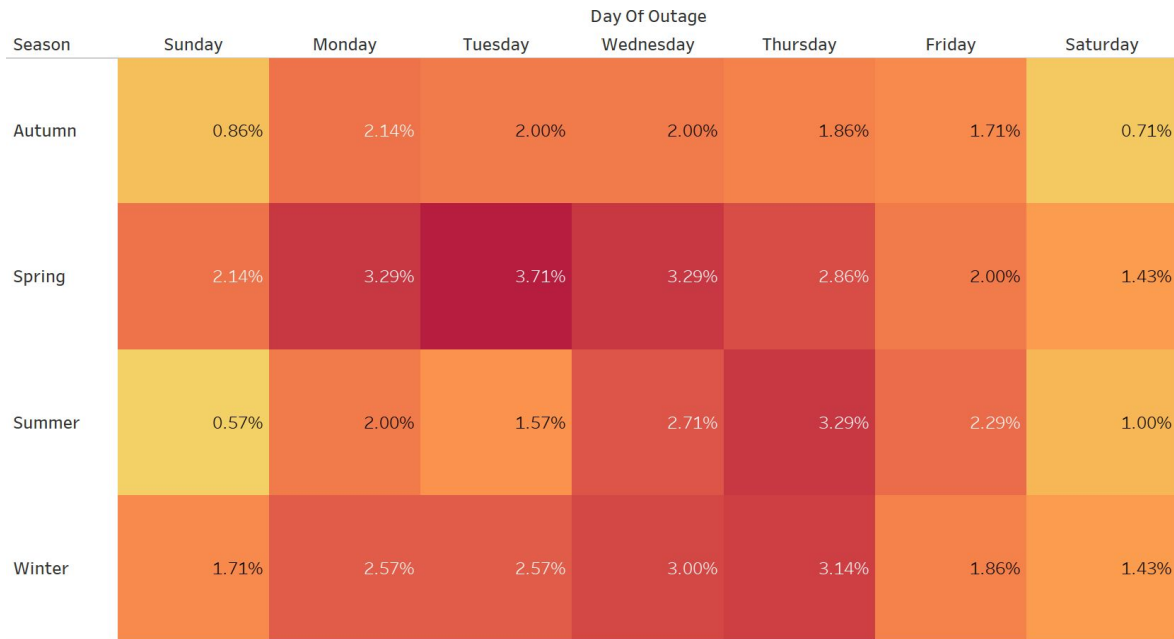


Monthly



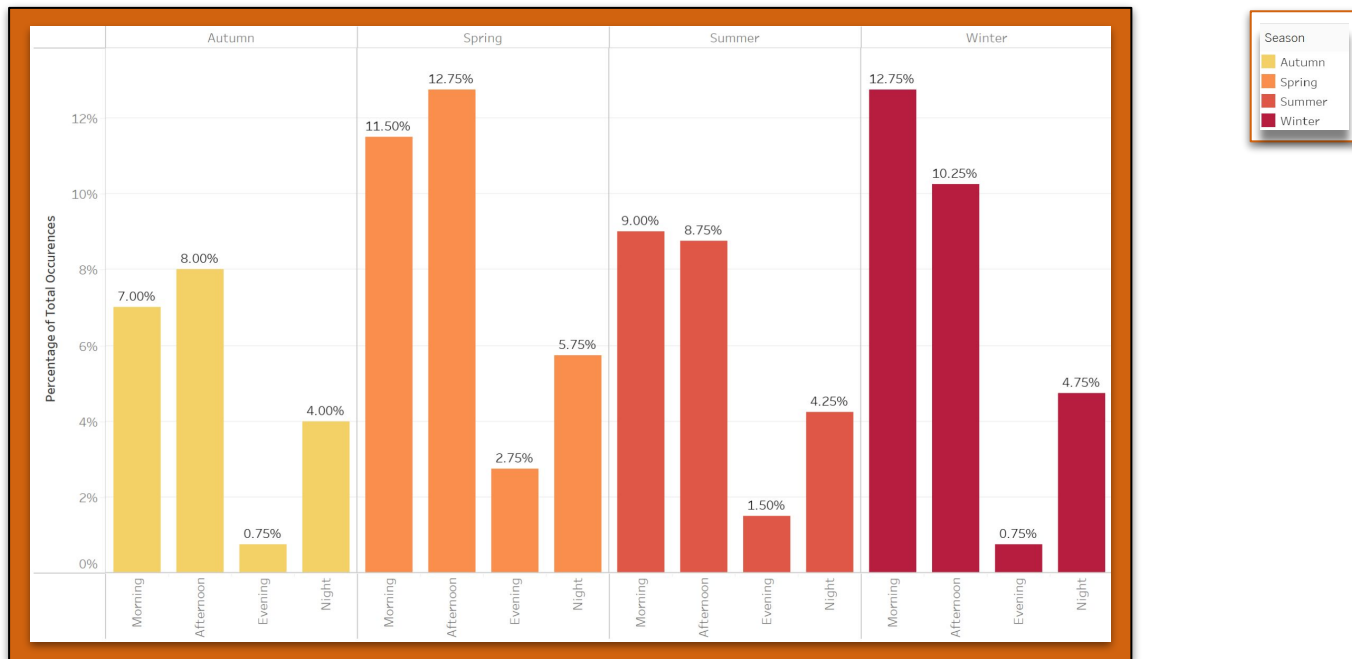


Percentage of Outages Throughout a Week for Each Season





Percentage of Power Outages Throughout a Day for Each Season



Key Insights

Severity = Outage Duration
Risk = Number of Outages

Most Severe,
Highest Risk



- **Longest Outages**
- **Most Outages**

Highest Risk Months

→ April & May

Highest Risk Days

→ Mon - Thurs

Highest Risk Time of Day

→ Morning - Afternoon

Most Severe Month

→ April

Moderate Severity,
High Risk



- **Moderate Outage Duration**
- **High Amount of Outages**

Highest Risk Months

→ January to February

Highest Risk Days

→ Mon - Thurs

Highest Risk Time of Day

→ Morning - Afternoon

Most Severe Month

→ December

Low Severity,
Moderate Risk



- **Low Outage Duration**
- **Moderate Amounts of Outages**

Highest Risk Months

→ June - July

Highest Risk Days

→ Wed - Fri

Highest Risk Time of Day

→ Morning - Afternoon

Most Severe Month

→ July

Low Severity,
Low Risk



- **Low Outage Duration**
- **Least Outages**

Highest Risk Months

→ Aug & Oct

Highest Risk Days

→ Mon - Wed

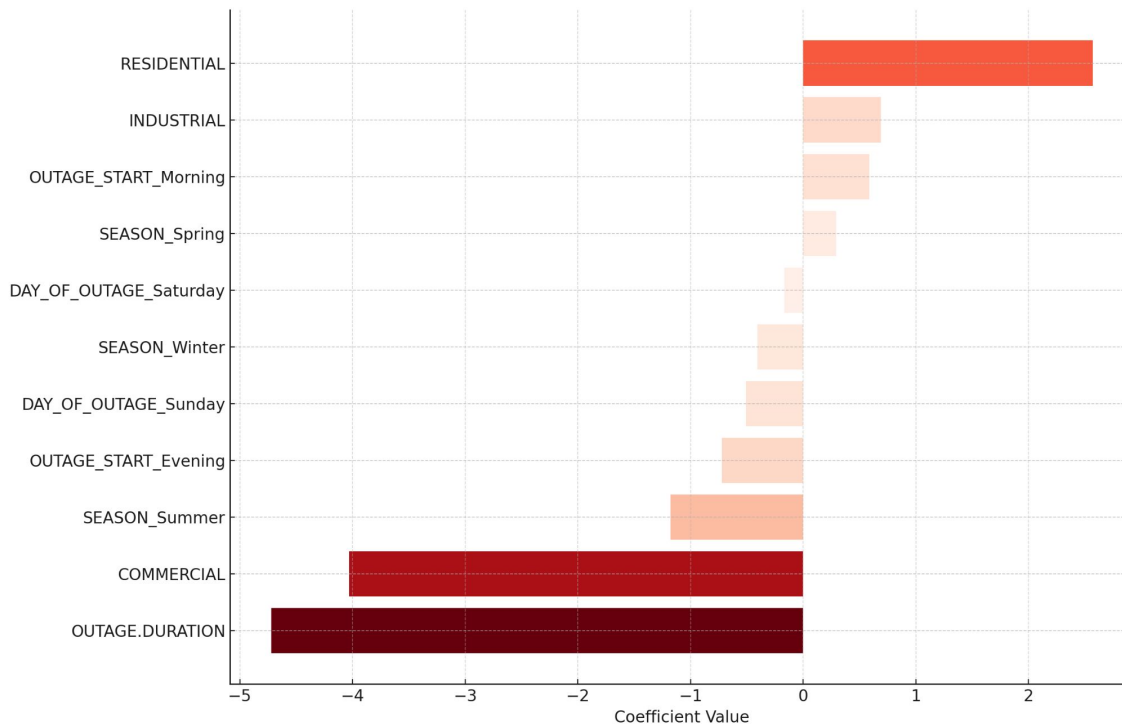
Highest Risk Time of Day

→ Morning - Afternoon

Most Severe Month

→ December

Logistic Regression Model - Coefficients



intercept
-2.1151312129716024

82%
accuracy

Demo

A power outage has occurred and you want to know if it is an intentional attack

Features

- Outage Day - Morning
 - YES (1)
- Residential
 - 2.3
- Commercial
 - -0.15
- Industrial
 - -0.15
- Outage Duration
 - -0.5
- Spring
 - YES (1)
- Intercept
 - -2.115

Calculation

$$\hat{y} = \frac{1}{1 + e^{-(2.1151312129716024 + 0.586934 + 5.90453 + 0.292132 + 2.360861 + 0.0225 + 0.0225)}}$$

=

0.99916

=

There is a **99.9% chance** this power outage was caused by an **intentional attack**

Action Plans

- **Quickly** inform law enforcement & authorities
- **Rapid** resource deployment
- **Swiftly** notify citizens of safety concerns





Intentional Attacks Dashboard

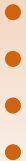
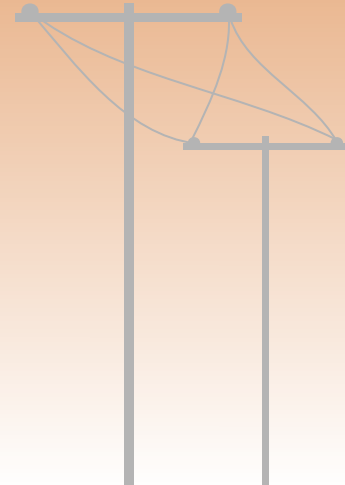


Click to
view our
dashboard



05

Predicting Power Outage Duration



Why This is Important

Prevention

- **Adjusts over time** based on new data
 - See current factors that cause the long power outages
 - Understand how to prevent long outages based on those factors

Rapid Response

- **Identify** longer outages and allocate resources quickly
 - **Prioritize** critical outages for immediate action
- Quickly inform **customers** how long the outage will last





How Can We Predict Power Outage Duration?

Multiple Linear Regression Model

What does it do?

Predicts how long an outage will last based on a power outage's attributes

How does it work?

Each **unique attribute for each factor** is given a **coefficient** that shows how much it affects the outage time.

The **coefficients are** combined and multiplied by the values of the attributes to give an estimated duration of the power outage

Dependent Variable (Response Variable)

Independent Variables (Predictors)

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \varepsilon$$

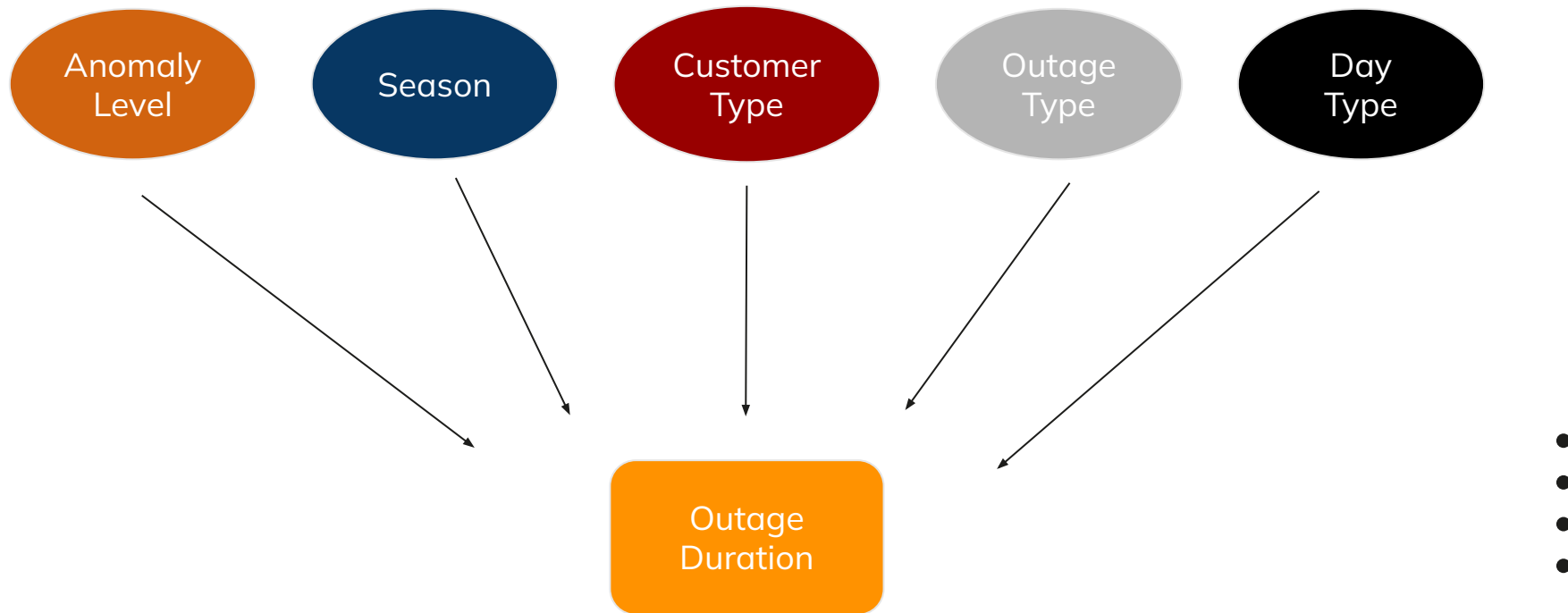
Y intercept

Slope Coefficient

Error Term

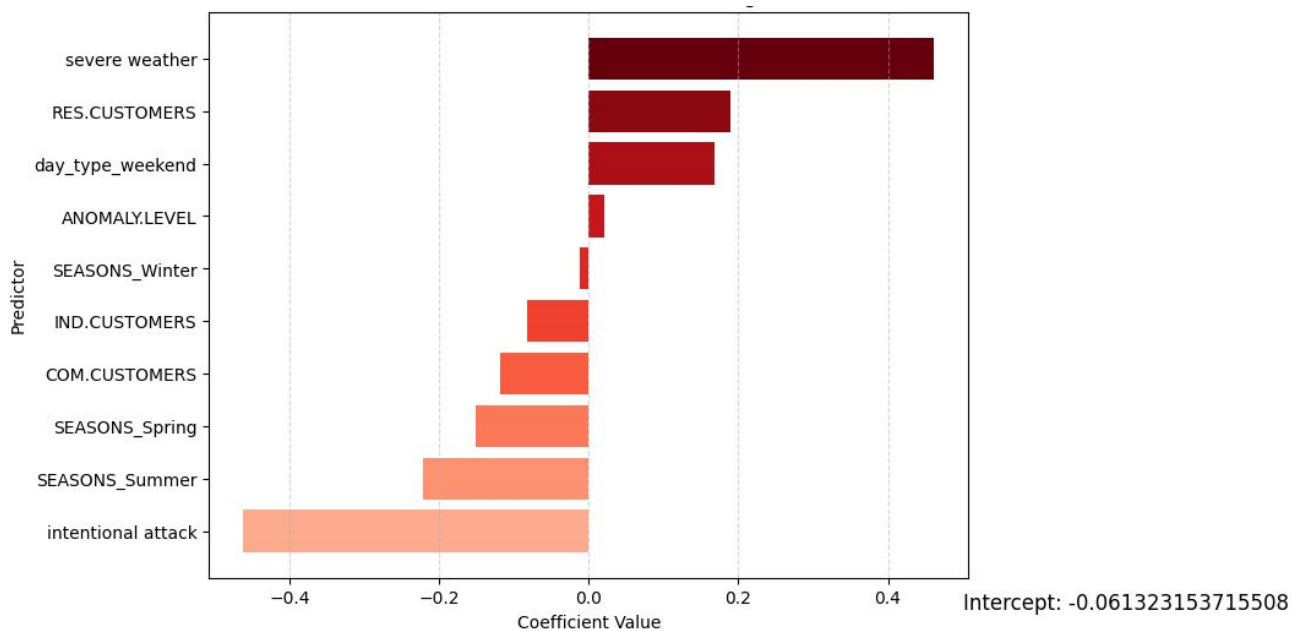


Our Factors





MLR Model - Coefficients



- Coefficients are Normalized
 - Range is Between -3 and 3



Demo

A power outage has occurred and you want to know whether it will be short or long

Features

- Intentional Attack
 - YES (1)
- Anomaly Level
 - -0.1
- Number of Commercial Customers in the Area
 - 0.5
- Number of Residential Customers in the Area
 - -0.2
- Number of Industrial Customers in the Area
 - 0.1
- Spring
 - YES (1)
- Weekend
 - YES (1)

Calculation

$$y = (1 \times -0.461527) + (-0.1 \times 0.020233) + (0.5 \times -0.1181) + (-0.2 \times 0.189174) + (0.1 \times -0.083097) + (1 \times -0.151216) + (1 \times 0.168216) + -0.061323153715507714$$

=

-0.614

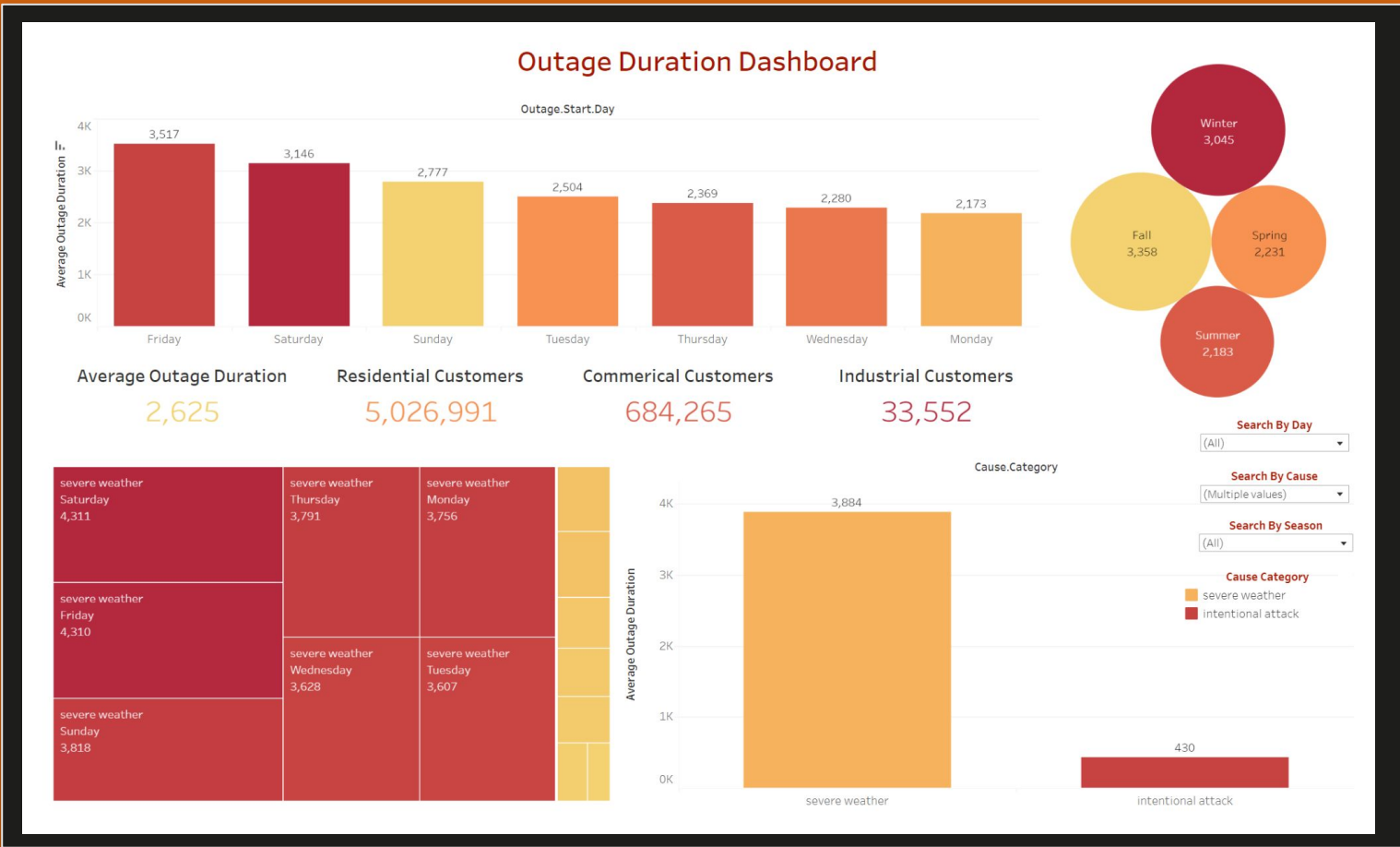
Action Plans

- **Reassure** customers that it will be a short outage
- **Prioritize** longer, more serious outages
- **Quickly identify** features for rapid resource deployment





Click to
view our
dashboard





Key Insights

Cause Category



Severe
Weather



Biggest impact on
power outage time

Day of the Week



Weekends



Friday, Saturday, and
Sunday have the
longest outages

Season



Fall

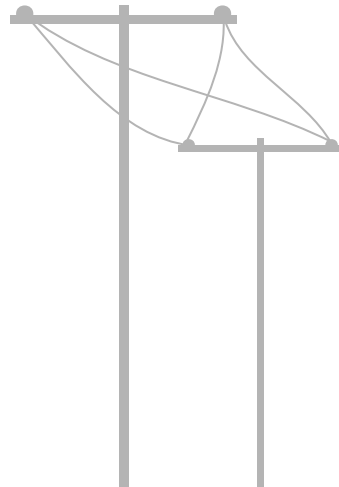


Power outages are
longest in the Fall



06

Recommendations





Outage Duration



Weather Resistance

Reinforce infrastructure to protect against severe weather.

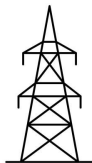
Weekend Maintenance

On weekends, maintenance will stay ready for repairs during outages.



Annual System Upgrades

Upgrade/upkeep infrastructure in preparation for Fall outages.



Surround Residential Areas

Maintenance will reside near residential areas to stay ready for outages.



Severe Weather



Upgrade Equipment

Prevent outages from outdated or eroded equipment.

Maintain Staffing

Keep enough staff on hand during high risk time to respond to potential outages quickly.

Monitor Conditions

Use forecasted measurements to predict when weather events may impact equipment, and prepare appropriately.

Seasonal Testing

Continuously test the resilience of equipment and response times of crews to reduce potential outage time as much as possible.





Intentional Attacks



Seasons to Watch Out For

Spring

Winter

Month to Watch Out For

Increase vigilance in
April

Time of Day to Watch Out For

Mornings and
Afternoons

Least Susceptible Months

Fall





Thank you for listening!





View our Project Code Here:

Logistic Regression - **Severe Weather Code**

Logistic Regression - **Intentional Attacks**

Multiple Linear Regression - **Outage Duration Code**



Click each one to view our entire
code





Works Cited

"Bloom Energy Outage Map." Bloom Energy, Bloom Energy Corporation, accessed April 21, 2024, <https://www.bloomenergy.com/bloom-energy-outage-map/>.

Morehouse, Catherine. "Power Grid Attacks Surge to New Peak." POLITICO, POLITICO, 10 Sept. 2023, www.politico.com/news/2023/09/10/power-grid-attacks-00114563.

"Surging Weather-related Power Outages." Climate Central, 13 Sept. 2022, climatecentral.org/climate-matters/surging-weather-related-power-outages.

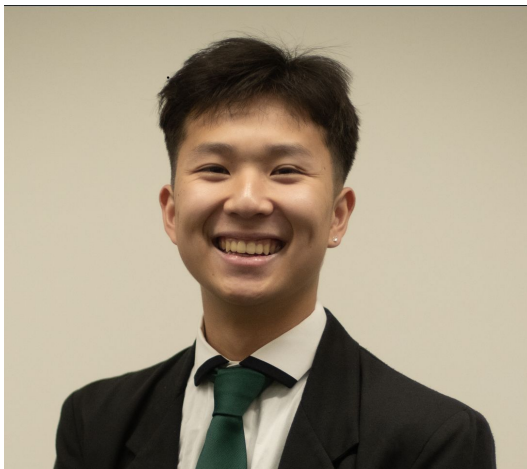


Meet the Team!





Dylan Ton



Project Manager

- 3rd year Computer Information Systems Major
 - Incoming Product Analytics Intern @ MyFitnessPal

Career goals:

- Work at a FAANG Company in the Data Science Field





Kevin Yuen



Data Analyst

- 3rd year Computer Information Systems Major
- 2023-2024 Member of Data Analytics @ MISSA
- Incoming Director of Membership @ MISSA

Career goals:

- Become a Data Analyst or Flight Analyst





Samyam Pyakurel



Data Analyst

- 4th year Computer Information Systems
- 2023-2024 Member of Data Analytics @ MISSA
- Process Improvement Specialist @ ASI CPP
- Platoon Leader @ United States Marine Corps Reserve

Career goals:

- Become a Data Analyst for a Government contractor company





Larissa Domingo

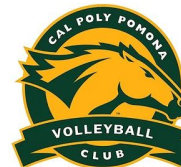


Data Analyst

- 4th year Computer Information Systems and Marketing Major
- 2023-2024 Member of Data Analytics @ MISSA
- 2022-2024 Secretary @ CPP Volleyball Sports Club
- Social Media and Marketing Assistant @ ASI CPP

Career goals:

- Become a Sports Data Analyst for a major professional sports team





Diego Cabral



Data Analyst

- 4th year Information Security & Forensics Major
- 2023-2024 Member of Data Analytics @ MISSA
- Experienced Tax and Immigration Professional

Career goals:

- Network Engineer / Information Security Analyst





Supriya Siwakoti



Data Analyst

- 4th year Computer Information Systems Major
- 2023-2024 Member of Data Analytics @ MISSA
- Experienced as a Tax Professional

Career goals:

- Become a Data Analyst





Nikhitha Vasiraju



Data Analyst

- 1st year Computer Science major
- 2023-2024 Member of Data Analytics @ MISSA

Career goals:

- Become a Data Analyst

